

## Qualification Pack

# Embedded System Junior Developer (O-Level Embedded System Design)

QP Code: NIE/ELE/Q0203

Version: 1.0

NSQF Level: 4

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## Qualification Pack

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## Qualification Pack

# NIE/ELE/Q0203: Embedded System Junior Developer (O-Level Embedded System Design)

## Brief Job Description

This course offers practical training in integrating multiple disciplines, such as engineering and computer science, to design and develop complex systems. It emphasizes hands-on experience and industry-relevant skills to prepare students for Embedded System analysis, design, and implementation roles. O Level in Embedded System Design aims to equip students with the practical skills and theoretical knowledge necessary to tackle real-world challenges in designing and implementing complex systems.

## Personal Attributes

A Junior Embedded System Developer should possess strong problem-solving skills and an analytical mindset to address real-time system challenges. They must demonstrate attention to detail, especially when coding and integrating hardware components. Effective communication and collaboration skills are essential for working within development teams and documenting technical processes. Adaptability and a willingness to continuously learn and apply new technologies are crucial in the rapidly evolving field of embedded systems.

## Applicable National Occupational Standards (NOS)

### Compulsory NOS:

1. [NIE/ELE/N0206: Embedded System Design & C- Programming](#)
2. [NIE/ELE/N0207: Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming](#)
3. [NIE/ELE/N0208: Sensor data acquisition and actuation in embedded systems](#)
4. [NIE/ELE/N0209: Linux Internals](#)
5. [NIE/ELE/N0220: Implementation of an Embedded System Project using ARM Cortex-M Series, Sensor Data Acquisition, and Linux Internals](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

## Qualification Pack (QP) Parameters

|                   |                                     |
|-------------------|-------------------------------------|
| <b>Sector</b>     | Electronics                         |
| <b>Sub-Sector</b> | Integrated Electronics and Circuits |

### Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Occupation</b>                                         | Embedded System Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Country</b>                                            | India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>NSQF Level</b>                                         | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Credits</b>                                            | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Minimum Educational Qualification &amp; Experience</b> | <p>Previous relevant Qualification of NSQF Level (NSQF Level 3.5 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 1.5 years of experience</p> <p>OR</p> <p>Previous relevant Qualification of NSQF Level (NSQF Level 3 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 3 Years of experience</p> <p>OR</p> <p>12th grade Pass (or equivalent in Science with Physics and Maths ) with NA of experience</p> <p>OR</p> <p>Completed 2nd year of the 3-year diploma after 10 (in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches ) with NA of experience</p> |
| <b>Minimum Level of Education for Training in School</b>  | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Next Review Date</b>                                   | 30/05/2027                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>NSQC Approval Date</b>                                 | 30/05/2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Reference code on NQR</b>                              | QG-04-EH-02591-2024-V1-NIELIT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>NQR Version</b>                                        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Qualification Pack

### NIE/ELE/N0206: Embedded System Design & C- Programming

#### Description

This NOS equips the students to explore the characteristics and components of embedded systems, setting up the development environment, and mastering C language syntax and basic programming constructs

#### Scope

The scope covers the following :

- The scope of this course encompasses an in-depth study of embedded systems and their characteristics, focusing on the C programming language tailored for embedded applications. Students will learn to set up the development environment and master C language syntax, variables, data types, control flow, pointers, and memory management techniques. The course also covers input/output operations, bit-level manipulation, peripheral register control, interrupt handling, and timer configurations. This comprehensive approach ensures students gain practical skills and theoretical knowledge essential for embedded systems programming.

#### Elements and Performance Criteria

##### *Embedded Systems and C Programming*

To be competent, the user/individual on the job must be able to:

- PC1.** Comprehend the fundamental characteristics and applications of embedded systems
- PC2.** Gain proficiency in the basics of the C programming language tailored for embedded systems
- PC3.** Successfully set up and configure an embedded systems development environment, including necessary tools and software

##### *Programming Constructs and Memory Management*

To be competent, the user/individual on the job must be able to:

- PC4.** Demonstrate understanding of C language syntax, including variables, data types, operators, and control flow structures
- PC5.** Apply pointers in various programming scenarios, understanding their significance in embedded systems
- PC6.** Efficiently use dynamic memory allocation and deallocation techniques, and implement memory management strategies for optimal performance in embedded systems

##### *Peripheral Control and Interrupt Handling*

To be competent, the user/individual on the job must be able to:

- PC7.** Access and manipulate peripheral registers to control hardware components
- PC8.** Implement interrupt handling mechanisms and develop interrupt service routines for responsive embedded applications
- PC9.** Configure timers and manage interrupt-based timer operations to achieve precise timing and scheduling in embedded systems

#### Knowledge and Understanding (KU)

## Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Know the steps and tools required to set up a development environment for embedded systems programming, including IDEs, compilers, and debuggers.
- KU2.** Gain in-depth knowledge of variables, data types, operators, control flow structures (if, switch, loops), and their application in embedded systems programming
- KU3.** Learn the configurations and operations of timers in embedded systems, including how to manage interrupts and implement interrupt-based timing mechanisms

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop the ability to identify, analyze, and solve problems in embedded systems by using debugging tools and techniques effectively.
- GS2.** Enhance communication and teamwork skills by collaborating with peers on projects, sharing knowledge, and providing constructive feedback
- GS3.** Foster adaptability by staying current with new tools, technologies, and methodologies in the rapidly evolving field of embedded systems

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Embedded Systems and C Programming</i>                                                                                                                                 | <b>30</b>    | <b>25</b>       | -             | <b>6</b>   |
| <b>PC1.</b> Comprehend the fundamental characteristics and applications of embedded systems                                                                               | -            | -               | -             | -          |
| <b>PC2.</b> Gain proficiency in the basics of the C programming language tailored for embedded systems                                                                    | -            | -               | -             | -          |
| <b>PC3.</b> Successfully set up and configure an embedded systems development environment, including necessary tools and software                                         | -            | -               | -             | -          |
| <i>Programming Constructs and Memory Management</i>                                                                                                                       | <b>30</b>    | <b>25</b>       | -             | <b>7</b>   |
| <b>PC4.</b> Demonstrate understanding of C language syntax, including variables, data types, operators, and control flow structures                                       | -            | -               | -             | -          |
| <b>PC5.</b> Apply pointers in various programming scenarios, understanding their significance in embedded systems                                                         | -            | -               | -             | -          |
| <b>PC6.</b> Efficiently use dynamic memory allocation and deallocation techniques, and implement memory management strategies for optimal performance in embedded systems | -            | -               | -             | -          |
| <i>Peripheral Control and Interrupt Handling</i>                                                                                                                          | <b>40</b>    | <b>30</b>       | -             | <b>7</b>   |
| <b>PC7.</b> Access and manipulate peripheral registers to control hardware components                                                                                     | -            | -               | -             | -          |
| <b>PC8.</b> Implement interrupt handling mechanisms and develop interrupt service routines for responsive embedded applications                                           | -            | -               | -             | -          |
| <b>PC9.</b> Configure timers and manage interrupt-based timer operations to achieve precise timing and scheduling in embedded systems                                     | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                          | <b>100</b>   | <b>80</b>       | -             | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                         |
|----------------------------|-----------------------------------------|
| <b>NOS Code</b>            | NIE/ELE/N0206                           |
| <b>NOS Name</b>            | Embedded System Design & C- Programming |
| <b>Sector</b>              | Electronics                             |
| <b>Sub-Sector</b>          |                                         |
| <b>Occupation</b>          | Embedded System Design                  |
| <b>NSQF Level</b>          | 4                                       |
| <b>Credits</b>             | 2                                       |
| <b>Version</b>             | 1.0                                     |
| <b>Last Reviewed Date</b>  | 30/05/2024                              |
| <b>Next Review Date</b>    | 30/05/2027                              |
| <b>NSQC Clearance Date</b> | 30/05/2024                              |



## Qualification Pack

### NIE/ELE/N0207: Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming

#### Description

This NOS equips the students to have the skills necessary to develop, optimize, and manage embedded applications, ensuring efficient and effective use of microcontroller resources.

#### Scope

The scope covers the following :

- This NOS provides a comprehensive understanding of Embedded C programming, embedded systems concepts, and ARM Cortex microcontroller architectures. Students will learn the fundamentals and advanced concepts of C programming, including memory management, pointers, and file operations. They will gain insights into the application areas, architecture, and development tools for embedded systems, with a focus on recent trends. The course covers ARM Cortex architectures in depth, including the M4 and A processors, and their development environments.

#### Elements and Performance Criteria

##### *Embedded C Programming*

To be competent, the user/individual on the job must be able to:

- PC1.** ● Demonstrate proficiency in writing and understanding complex C programs involving control flow, data structures, and memory management.
- PC2.** ● Implement algorithms using pointers and understand their role in memory manipulation and efficient programming.
- PC3.** ● Develop error-free programs with correct usage of bitwise operators and typecasting for optimized code execution

##### *ARM Cortex Architecture*

To be competent, the user/individual on the job must be able to:

- PC4.** ● Demonstrate the ability to configure and utilize Cortex M4 peripherals effectively in embedded applications, including GPIO, timers, UARTs, and ADCs
- PC5.** ● Develop and debug assembly language programs for ARM Cortex-M4 processors, ensuring proper memory access and interrupt handling
- PC6.** ● Design and implement interrupt service routines (ISRs) for real-time event management on Cortex M4 microcontrollers

##### *Power Management in Embedded Systems*

To be competent, the user/individual on the job must be able to:

- PC7.** ● Design power optimization strategies using various power modes like Active, Sleep, and Standby to extend battery life and improve energy efficiency
- PC8.** ● Implement clock management techniques, including clock gating and dynamic frequency scaling, to minimize power consumption during idle periods
- PC9.** ● Develop energy profiling techniques and utilize power management APIs to optimize energy usage and enhance system performance

## Qualification Pack

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Gain knowledge of advanced topics such as conditional compilation, pre-processor directives, and file operations in C programming
- KU2.** Comprehend the architecture and organization of ARM Cortex processors, including the Cortex-M4 register set, instruction set architecture (ISA), and memory hierarchy
- KU3.** Understand the principles of power management in embedded systems, including power modes and clock management techniques

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop the ability to analyze complex problems in embedded systems programming and find efficient solution using C programming, ARM Cortex architecture, and power management strategies
- GS2.** Enhance critical thinking skills by evaluating different approaches to memory management, peripheral control, and power optimization in embedded systems development
- GS3.** Improve communication skills by documenting code, explaining programming concepts, and presenting project findings effectively to peers and stakeholders in the field of embedded systems and microcontroller programming

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Embedded C Programming</i>                                                                                                                                      | <b>30</b>    | <b>25</b>       | -             | <b>6</b>   |
| <b>PC1.</b> ● Demonstrate proficiency in writing and understanding complex C programs involving control flow, data structures, and memory management.              | -            | -               | -             | -          |
| <b>PC2.</b> ● Implement algorithms using pointers and understand their role in memory manipulation and efficient programming.                                      | -            | -               | -             | -          |
| <b>PC3.</b> ● Develop error-free programs with correct usage of bitwise operators and typecasting for optimized code execution                                     | -            | -               | -             | -          |
| <i>ARM Cortex Architecture</i>                                                                                                                                     | <b>30</b>    | <b>25</b>       | -             | <b>7</b>   |
| <b>PC4.</b> ● Demonstrate the ability to configure and utilize Cortex M4 peripherals effectively in embedded applications, including GPIO, timers, UARTs, and ADCs | -            | -               | -             | -          |
| <b>PC5.</b> ● Develop and debug assembly language programs for ARM Cortex-M4 processors, ensuring proper memory access and interrupt handling                      | -            | -               | -             | -          |
| <b>PC6.</b> ● Design and implement interrupt service routines (ISRs) for real-time event management on Cortex M4 microcontrollers                                  | -            | -               | -             | -          |
| <i>Power Management in Embedded Systems</i>                                                                                                                        | <b>40</b>    | <b>30</b>       | -             | <b>7</b>   |
| <b>PC7.</b> ● Design power optimization strategies using various power modes like Active, Sleep, and Standby to extend battery life and improve energy efficiency  | -            | -               | -             | -          |
| <b>PC8.</b> ● Implement clock management techniques, including clock gating and dynamic frequency scaling, to minimize power consumption during idle periods       | -            | -               | -             | -          |
| <b>PC9.</b> ● Develop energy profiling techniques and utilize power management APIs to optimize energy usage and enhance system performance                        | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------|--------------|-----------------|---------------|------------|
| <b>NOS Total</b>                 | <b>100</b>   | <b>80</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                              |
|----------------------------|------------------------------------------------------------------------------|
| <b>NOS Code</b>            | NIE/ELE/N0207                                                                |
| <b>NOS Name</b>            | Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming |
| <b>Sector</b>              | Electronics                                                                  |
| <b>Sub-Sector</b>          |                                                                              |
| <b>Occupation</b>          | Embedded System Design                                                       |
| <b>NSQF Level</b>          | 4                                                                            |
| <b>Credits</b>             | 2                                                                            |
| <b>Version</b>             | 1.0                                                                          |
| <b>Last Reviewed Date</b>  | 30/05/2024                                                                   |
| <b>Next Review Date</b>    | 30/05/2027                                                                   |
| <b>NSQC Clearance Date</b> | 30/05/2024                                                                   |

## Qualification Pack

### NIE/ELE/N0208: Sensor data acquisition and actuation in embedded systems

#### Description

The objective is to understand transducers' role, sensor classifications, principles of transduction, and sensor characteristics

#### Scope

The scope covers the following :

- This course offers an in-depth introduction to transducers, sensors, and actuators, focusing on their roles in measurement and control systems. Students will learn about the principles of transduction, sensor and actuator classifications, characteristics, and applications in various fields. The course also covers interfacing these components with microcontrollers, signal conditioning, and calibration techniques. By the end, students will be equipped to implement sensors and actuators in real-world applications, enhancing their skills in modern measurement and control technologies

#### Elements and Performance Criteria

##### *Sensor Fundamentals and Types*

To be competent, the user/individual on the job must be able to:

- PC1.** ● Ability to explain the basic principles of sensors and their functions accurately
- PC2.** ● Demonstrate proficiency in classifying sensors based on measured quantities and sensing mechanisms
- PC3.** ● Evaluate sensor characteristics such as sensitivity, accuracy, linearity, and range in practical scenarios

##### *Actuator Fundamentals and Applications*

To be competent, the user/individual on the job must be able to:

- PC4.** ● Demonstrate understanding of actuator functions and classify them based on output types and characteristics
- PC5.** ● Apply knowledge of actuators to solve real-world problems in robotics, automation, and fluid control
- PC6.** ● Evaluate and select appropriate actuators for specific applications based on performance criteria like response time, force, torque, and displacement

##### *Integration and Interfacing Techniques*

To be competent, the user/individual on the job must be able to:

- PC7.** ● Successfully interface sensors and actuators with microcontrollers and embedded systems
- PC8.** ● Apply signal conditioning and amplification techniques effectively to improve sensor and actuator performance
- PC9.** ● Demonstrate calibration and compensation methods to ensure accurate integration and reliable functioning in control systems

#### Knowledge and Understanding (KU)

## Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Understand the role of transducers in measurement and control systems
- KU2.** Understanding Actuators and their Applications-Comprehend the functions and classifications of actuators according to their output types and characteristics
- KU3.** Understand the integration of sensors and actuators with microcontrollers and embedded systems

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze sensor and actuator requirements in different applications to propose suitable solutions.
- GS2.** Conduct experiments to validate sensor accuracy, linearity, and range under different conditions
- GS3.** Effectively communicate technical concepts related to transducers, sensors, and actuators

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Sensor Fundamentals and Types</i>                                                                                                                                | <b>30</b>    | <b>25</b>       | -             | <b>6</b>   |
| <b>PC1.</b> ● Ability to explain the basic principles of sensors and their functions accurately                                                                     | -            | -               | -             | -          |
| <b>PC2.</b> ● Demonstrate proficiency in classifying sensors based on measured quantities and sensing mechanisms                                                    | -            | -               | -             | -          |
| <b>PC3.</b> ● Evaluate sensor characteristics such as sensitivity, accuracy, linearity, and range in practical scenarios                                            | -            | -               | -             | -          |
| <i>Actuator Fundamentals and Applications</i>                                                                                                                       | <b>30</b>    | <b>25</b>       | -             | <b>7</b>   |
| <b>PC4.</b> ● Demonstrate understanding of actuator functions and classify them based on output types and characteristics                                           | -            | -               | -             | -          |
| <b>PC5.</b> ● Apply knowledge of actuators to solve real-world problems in robotics, automation, and fluid control                                                  | -            | -               | -             | -          |
| <b>PC6.</b> ● Evaluate and select appropriate actuators for specific applications based on performance criteria like response time, force, torque, and displacement | -            | -               | -             | -          |
| <i>Integration and Interfacing Techniques</i>                                                                                                                       | <b>40</b>    | <b>30</b>       | -             | <b>7</b>   |
| <b>PC7.</b> ● Successfully interface sensors and actuators with microcontrollers and embedded systems                                                               | -            | -               | -             | -          |
| <b>PC8.</b> ● Apply signal conditioning and amplification techniques effectively to improve sensor and actuator performance                                         | -            | -               | -             | -          |
| <b>PC9.</b> ● Demonstrate calibration and compensation methods to ensure accurate integration and reliable functioning in control systems                           | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                    | <b>100</b>   | <b>80</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| <b>NOS Code</b>            | NIE/ELE/N0208                                             |
| <b>NOS Name</b>            | Sensor data acquisition and actuation in embedded systems |
| <b>Sector</b>              | Electronics                                               |
| <b>Sub-Sector</b>          |                                                           |
| <b>Occupation</b>          | Embedded System Design                                    |
| <b>NSQF Level</b>          | 4                                                         |
| <b>Credits</b>             | 2                                                         |
| <b>Version</b>             | 1.0                                                       |
| <b>Last Reviewed Date</b>  | 30/05/2024                                                |
| <b>Next Review Date</b>    | 30/05/2027                                                |
| <b>NSQC Clearance Date</b> | 30/05/2024                                                |

## Qualification Pack

### NIE/ELE/N0209: Linux Internals

#### Description

This Course is to provide a comprehensive understanding of Embedded Linux, focusing on its architecture, kernel basics, system calls, and file and process management. Students will learn about memory and device management, inter-process communication, multi-threading, synchronization, and networking concepts specific to embedded systems.

#### Scope

The scope covers the following :

- This course encompasses the fundamental principles and practical applications of Embedded Linux. It aims to provide students with a thorough understanding of how Linux is integrated into embedded systems. The course covers key aspects of Linux kernel architecture, system calls, file and process management, memory and device management, inter-process communication, multi-threading, synchronization, and networking in embedded environments. By the end of the course, students will be proficient in the concepts and practices necessary for developing and managing embedded systems using Linux.

#### Elements and Performance Criteria

##### *Embedded Linux Architecture and Components*

To be competent, the user/individual on the job must be able to:

- PC1.** ● Students should be able to explain the overall structure of Embedded Linux and identify its key components
- PC2.** ● Students should accurately describe the differences between Embedded Linux and standard Linux distributions
- PC3.** ● Analyze and discuss case studies where Embedded Linux is used, highlighting its advantages and challenges

##### *Linux Kernel Basics*

To be competent, the user/individual on the job must be able to:

- PC4.** ● Students should demonstrate how the kernel manages resources such as processes, memory, and devices
- PC5.** ● Perform practical exercises that involve configuring and optimizing kernel parameters for resource management
- PC6.** ● Write simple programs that make use of system calls to perform basic tasks like file operations or process management

##### *Inter-process Communication and Multi-threading*

To be competent, the user/individual on the job must be able to:

- PC7.** ● Write and test programs that utilize IPC to communicate between processes
- PC8.** ● Develop and debug multi-threaded applications, ensuring proper synchronization and resource sharing

## Qualification Pack

- PC9.** ● Analyze and present case studies that involve complex IPC and multi-threading scenarios, discussing the solutions and their impact on system performance

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the hierarchical structure of Embedded Linux, including the kernel, libraries, and application layers
- KU2.** Familiarity with the roles and functions of each component within the Embedded Linux architecture
- KU3.** Comprehensive understanding of the Linux kernel architecture, including its layers and modules
- KU4.** Knowledge of how different parts of the kernel interact and function together
- KU5.** Comprehensive knowledge of multi-threading concepts and techniques in Linux
- KU6.** Skills to develop, debug, and optimize multi-threaded applications

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Skills in presenting technical concepts and solutions to diverse audiences, including non-technical stakeholders
- GS2.** Strong ability to identify, diagnose, and resolve issues within embedded Linux systems.
- GS3.** Proficiency in using debugging tools and techniques to fix kernel, application, and system-level problems.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Embedded Linux Architecture and Components</i>                                                                                                                      | <b>30</b>    | <b>25</b>       | -             | <b>6</b>   |
| <b>PC1.</b> ● Students should be able to explain the overall structure of Embedded Linux and identify its key components                                               | -            | -               | -             | -          |
| <b>PC2.</b> ● Students should accurately describe the differences between Embedded Linux and standard Linux distributions                                              | -            | -               | -             | -          |
| <b>PC3.</b> ● Analyze and discuss case studies where Embedded Linux is used, highlighting its advantages and challenges                                                | -            | -               | -             | -          |
| <i>Linux Kernel Basics</i>                                                                                                                                             | <b>30</b>    | <b>25</b>       | -             | <b>7</b>   |
| <b>PC4.</b> ● Students should demonstrate how the kernel manages resources such as processes, memory, and devices                                                      | -            | -               | -             | -          |
| <b>PC5.</b> ● Perform practical exercises that involve configuring and optimizing kernel parameters for resource management                                            | -            | -               | -             | -          |
| <b>PC6.</b> ● Write simple programs that make use of system calls to perform basic tasks like file operations or process management                                    | -            | -               | -             | -          |
| <i>Inter-process Communication and Multi-threading</i>                                                                                                                 | <b>40</b>    | <b>30</b>       | -             | <b>7</b>   |
| <b>PC7.</b> ● Write and test programs that utilize IPC to communicate between processes                                                                                | -            | -               | -             | -          |
| <b>PC8.</b> ● Develop and debug multi-threaded applications, ensuring proper synchronization and resource sharing                                                      | -            | -               | -             | -          |
| <b>PC9.</b> ● Analyze and present case studies that involve complex IPC and multi-threading scenarios, discussing the solutions and their impact on system performance | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                       | <b>100</b>   | <b>80</b>       | -             | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                        |
|----------------------------|------------------------|
| <b>NOS Code</b>            | NIE/ELE/N0209          |
| <b>NOS Name</b>            | Linux Internals        |
| <b>Sector</b>              | Electronics            |
| <b>Sub-Sector</b>          |                        |
| <b>Occupation</b>          | Embedded System Design |
| <b>NSQF Level</b>          | 4                      |
| <b>Credits</b>             | 2                      |
| <b>Version</b>             | 1.0                    |
| <b>Last Reviewed Date</b>  | 30/05/2024             |
| <b>Next Review Date</b>    | 30/05/2027             |
| <b>NSQC Clearance Date</b> | 30/05/2024             |

## Qualification Pack

# NIE/ELE/N0220: Implementation of an Embedded System Project using ARM Cortex-M Series, Sensor Data Acquisition, and Linux Internals

## Description

This NOS focuses on developing an embedded system using ARM Cortex-M series microcontrollers for sensor data acquisition and processing. Students will integrate various sensors with the microcontroller, leveraging real-time data handling techniques. This NOS also covers Linux Internals, providing insights into kernel configuration, memory management, and device communication. Through hands-on implementation, participants will design efficient, real-world embedded solutions.

## Scope

The scope covers the following :

- The scope of this NOS includes designing an embedded system using ARM Cortex-M series microcontrollers for efficient sensor data acquisition and real-time processing. It involves integrating multiple sensors with the microcontroller and utilizing Linux Internals for tasks like kernel configuration and memory management. This NOS also focuses on optimizing communication between devices and the embedded system. Additionally, it provides hands-on experience in developing robust embedded applications.

## Elements and Performance Criteria

### *Implementation of an Embedded System Project using ARM Cortex-M Series, Sensor Data Acquisition, and Linux Internals*

To be competent, the user/individual on the job must be able to:

- PC1.** Ability to configure ARM Cortex-M peripherals (GPIO, timers, UART) effectively for real-time operations and communication.
- PC2.** Demonstrate proficiency in writing optimized embedded C code for microcontroller tasks, ensuring efficient performance
- PC3.** Effectively interface various sensors with the ARM Cortex-M microcontroller, ensuring accurate data acquisition.
- PC4.** Demonstrate the ability to process and respond to sensor inputs in real-time, meeting system performance requirements.
- PC5.** Ability to configure and compile the Linux kernel tailored to embedded systems, ensuring efficient use of resources
- PC6.** Demonstrate the ability to manage devices and peripherals using Linux Internals, facilitating seamless interaction in the embedded system.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the ARM Cortex-M architecture, including memory hierarchy, instruction sets, and peripheral interfacing.

## Qualification Pack

- KU2.** Knowledge of embedded C programming techniques for handling GPIOs, timers, interrupts, and low-level microcontroller tasks.
- KU3.** Knowledge of various sensors (e.g., temperature, pressure, motion) and their characteristics such as sensitivity, accuracy, and range.
- KU4.** Understanding techniques for acquiring and processing sensor data, including analog-to-digital conversion and signal conditioning.
- KU5.** Understanding the Linux kernel's structure, including process management, memory management, and device drivers for embedded systems.
- KU6.** Understanding how Linux handles memory, CPU resources, and processes, and applying this knowledge to optimize embedded system performance.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to identify, troubleshoot, and resolve issues related to microcontroller configuration and programming.
- GS2.** Competence in working with a team to integrate sensors with microcontrollers, ensuring smooth data flow and communication.
- GS3.** Ability to analyze sensor data requirements and design effective data acquisition strategies for accurate real-time processing.
- GS4.** Ability to evaluate system performance and optimize Linux kernel configurations for embedded applications.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Implementation of an Embedded System Project using ARM Cortex-M Series, Sensor Data Acquisition, and Linux Internals</i>                            | -            | -               | <b>80</b>     | <b>20</b>  |
| <b>PC1.</b> Ability to configure ARM Cortex-M peripherals (GPIO, timers, UART) effectively for real-time operations and communication.                 | -            | -               | -             | -          |
| <b>PC2.</b> Demonstrate proficiency in writing optimized embedded C code for microcontroller tasks, ensuring efficient performance                     | -            | -               | -             | -          |
| <b>PC3.</b> Effectively interface various sensors with the ARM Cortex-M microcontroller, ensuring accurate data acquisition.                           | -            | -               | -             | -          |
| <b>PC4.</b> Demonstrate the ability to process and respond to sensor inputs in real-time, meeting system performance requirements.                     | -            | -               | -             | -          |
| <b>PC5.</b> Ability to configure and compile the Linux kernel tailored to embedded systems, ensuring efficient use of resources                        | -            | -               | -             | -          |
| <b>PC6.</b> Demonstrate the ability to manage devices and peripherals using Linux Internals, facilitating seamless interaction in the embedded system. | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                       | -            | -               | <b>80</b>     | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | NIE/ELE/N0220                                                                                                        |
| <b>NOS Name</b>            | Implementation of an Embedded System Project using ARM Cortex-M Series, Sensor Data Acquisition, and Linux Internals |
| <b>Sector</b>              | Electronics                                                                                                          |
| <b>Sub-Sector</b>          |                                                                                                                      |
| <b>Occupation</b>          | Embedded System Design                                                                                               |
| <b>NSQF Level</b>          | 4                                                                                                                    |
| <b>Credits</b>             | 5                                                                                                                    |
| <b>Version</b>             | 1.0                                                                                                                  |
| <b>Last Reviewed Date</b>  | 30/05/2024                                                                                                           |
| <b>Next Review Date</b>    | 30/05/2027                                                                                                           |
| <b>NSQC Clearance Date</b> | 30/05/2024                                                                                                           |

## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

## Qualification Pack

**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings

## Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/03/2029                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria. The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns the roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds

## Qualification Pack

and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

### Minimum Aggregate Passing % at QP Level : 50

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage |
|--------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|-----------|
| NIE/ELE/N0206.Embedded System Design & C-Programming                                       | 100          | 80              | 0             | 20         | 200         | 21        |
| NIE/ELE/N0207.Microcontroller - ARM Cortex M series - Architecture, Peripheral Programming | 100          | 80              | 0             | 20         | 200         | 21        |

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| National Occupational Standards                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| NIE/ELE/N0208.Sensor data acquisition and actuation in embedded systems                                                            | 100          | 80              | 0             | 20         | 200         | 21         |
| NIE/ELE/N0209.Linux Internals                                                                                                      | 100          | 80              | 0             | 20         | 200         | 21         |
| NIE/ELE/N0220.Implementation of an Embedded System Project using ARM Cortex-M Series, Sensor Data Acquisition, and Linux Internals | 0            | 0               | 80            | 20         | 100         | 11         |
| DGT/VSQ/N0102.Employability Skills (60 Hours)                                                                                      | 20           | 30              | -             | -          | 50          | 5          |
| <b>Total</b>                                                                                                                       | <b>420</b>   | <b>350</b>      | <b>80</b>     | <b>100</b> | <b>950</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                      |
|-------------|------------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)                    |
| <b>NSQF</b> | National Skills Qualifications Framework             |
| <b>QP</b>   | Qualifications Pack                                  |
| <b>TVET</b> | Technical and Vocational Education and Training      |
| <b>AA</b>   | Assessment Agency                                    |
| <b>AB</b>   | Awarding Body                                        |
| <b>ISCO</b> | International Standard Classification of Occupations |
| <b>NCO</b>  | National Classification of Occupations               |
| <b>NCrF</b> | National Credit Framework                            |
| <b>NOS</b>  | National Occupational Standard(s)                    |
| <b>NQR</b>  | National Qualification Register                      |
| <b>NSQF</b> | National Skills Qualifications Framework             |
| <b>OJT</b>  | On the Job Training                                  |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |
| <b>Knowledge and Understanding (KU)</b>      | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                         |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |
| <b>National Occupational Standard</b>   | NOS defines the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.                                                                                                                                                                              |
| <b>Qualification</b>                    | A formal outcome of an assessment and validation process which is obtained when a the competent body determines that an individual has achieved learning outcomes to given standards                                                                                                                                                                                   |
| <b>Qualification File</b>               | A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.                                                                                                                                 |
| <b>Sector</b>                           | A grouping of professional activities on the basis of their main economic function, product, service or technology.                                                                                                                                                                                                                                                    |
| <b>Long Term Training</b>               | Long-term skilling means any vocational training program undertaken for a year and above. <a href="https://ncvet.gov.in/sites/default/files/NCVET.pdf">https://ncvet.gov.in/sites/default/files/NCVET.pdf</a>                                                                                                                                                          |